

Experimentally studying path-finding problem between conjugates in supersingular isogeny graphs: Optimizing primes and powers to speed-up cycle finding

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Abstract

We study the problem of finding paths between conjugate supersingular elliptic curves, a subroutine of cycle finding algorithm in the graph $\chi_l(\overline{\mathbb{F}_p})$ of all supersingular elliptic curves over $\mathbb{F}_{p^2}$, connected by l isogenies for primes l, p . We prove that asymptotically the most time-efficient way of traversing the graph to find conjugate paths is seeing $i(= 3)$ steps together, a question posed in Remark 3.5 by Eisenträger *et. al.* [EHL⁺20]. We outline procedures for choosing l which will accelerate the process of conjugate path-finding. We have experimentally investigated the paths between frobenius conjugates for wide ranges of small prime l , which adds to previous studies [ACNL⁺23]. Experiments confirm that selecting l, i optimally speeds up the cycle finding, in an analogous way as the equivalent problem of isogeny finding to subfield curve was hastened by the Supersolver method [CRSCS22]. We introduce sets to experimentally study the structure of the isogeny graphs when short cycles are present.

1 Introduction

The public key cryptosystems used today are based on integer factoring or the discrete logarithm problem in finite fields or elliptic curves. The presence of a quantum computing facility of a few thousand logical qubits would probably suffice to break these systems [RNSL17], given Shor’s algorithm [Sho99]. Post-quantum cryptography is the study of cryptosystems that remain secure even if a quantum computer is available.

Let p be a large prime defining a finite field $\mathbb{F}_p$. The supersingular isogeny graph $\chi_l(\overline{\mathbb{F}_p})$ for a prime $l(\neq p)$ is a directed graph with the vertices being the isomorphism classes of supersingular elliptic curves, and the directed edges being equivalent l isogenies where both the supersingular elliptic curves and isogenies are defined over $\overline{\mathbb{F}_p}$. Isogeny based cryptography is a candidate of post-quantum cryptography, and it relies on the hardness of various [EHL⁺20, DFG19, BKV19, DFKL⁺20, CLM⁺18] types of path-finding problem in supersingular isogeny graphs between same or different base and final curves. Endomorphisms being isogenies between same supersingular elliptic curves can be perceived as cycles within the graph $\chi_l(\overline{\mathbb{F}_p})$. The problem of finding paths in these graphs is equivalent to computing endomorphism rings [EHL⁺18] which can be reduced to computing one endomorphism [PW24, MW23]. Time complexity improvements in the path-finding problem are important from the viewpoint of choosing better cryptographic parameters [CRSCS22, BDF⁺24, BBC⁺21]. Computational observations [ACNL⁺23] reveal that for $l = 2, 3$, vertices are generally closer to their frobenius conjugates in

$\chi_l(\overline{\mathbb{F}_p})$, in the sense that if one performs a pseudo-random walk, lesser steps will be required when the base and final curves are frobenius conjugates of each other compared to the case when they are two arbitrary supersingular elliptic curves. The path-finding problem between conjugates is the problem of finding a path between a given supersingular elliptic curve E and its frobenius conjugate curve $E^{(p)}$. Paths between conjugate supersingular elliptic curves are used to construct cycles which aid in computing endomorphism rings [EHL⁺20]. Relying on efficient computation of cycles, Kohel [Koh96] first studied the problem of finding endomorphism rings, which was later continued by Galbraith *et. al.* [GPS20] along a different line. The runtime of Kohel’s algorithm [Koh96], which finds subring of finite index in the endomorphism ring is $O(p^{1+\epsilon})$ while [GPS20] calls cycle finding algorithm $O(\log p)$ times. The algorithm of Eisenträger *et. al.*, [EHL⁺20] is probabilistic and depends on certain heuristics and constructs two cycles to recover endomorphism ring from a Bass suborder in time $O((\log p)^2 p^{\frac{1}{2}})$. An essential component of the above algorithms is that they are all based on finding cycles. Recent endomorphism ring computation algorithms are also available [FIK⁺23, ES24, XZQ24]. Despite these newer algorithms which remove some conditions in [EHL⁺20], the algorithm in it remains relevant due to optimized complexity, especially as p increases.

The cycle computation problem is also important as an algorithmic number theory problem and its hardness can be utilized for devising secure cryptographic schemes. Important applications beside endomorphism ring computation [EHL⁺20] are in constructing collision resistant hash functions [CLG09], SQIsign [DFKL⁺20], etc.

1.1 Motivation and our contributions

In this section, we list our contributions(which we have itemized) and the motivations behind them. Our code to perform experiments is available at:

<https://github.com/Madhurima11/cycle-computation-practical-methods>

The path-finding between conjugates is a specialised case of the general path-finding problem between any two arbitrary supersingular elliptic curves. The connectedness of the graph $\chi_l(\overline{\mathbb{F}_p})$ implies that for large $i = O(\log p)$, isogenies between two given supersingular elliptic curves is guaranteed [Theorem 25], which is not the case for small i . Hence a relevant problem is that, given small i , enquiring existence of an isogeny of degree l^i between frobenius conjugates. The important aspect here is that when we are performing a psuedo-random walk, increasing i at each step means we can view a larger portion of the graph. This gives the advantage of choosing a direction that would help us to land with our desired vertex quickly. On the other hand, the cost of finding isogeny grows on increasing i .

- Inspired by remark 3.5 of [EHL⁺20] which says we can tinker their algorithm[Algorithm 3.4 [EHL⁺20]] with the cases when the distances between frobenius conjugates “in the graph is bounded by some fixed integer B ”, we prove that [Section 4.1] when solving the path finding problem between conjugates, taking $i = 3$ asymptotically balances the cost of finding isogenies with the advantage of broader view. We have described optimal methods for each bound up to 3. We achieve the same asymptotic cost of $O(l^2 \log p)$ as in [EHL⁺20], which is the most optimal till now, but with the advantage of practical improvements. Our methods adapt the Supersolver method [CRSCS22] of isogeny computation to the cycle computation scenario, and the experiments[Table 1] validate that it speeds the computation.

The subfield curve search, that is searching for a supersingular elliptic curve in $\mathbb{F}_p$, after initiating a pseudo-random walk in $\chi_l(\overline{\mathbb{F}_p})$ speeds up [CRSCS22] for appropriate l .

- Inspired by the above fact, we note[C,D] that similar acceleration occurs for conjugate path-finding. We can[Section 6.1] choose l with the goal of minimizing the total cost.

The above discussion makes it necessary to study the nature of the path between conjugates in $\chi_l(\overline{\mathbb{F}_p})$ as l varies. Experimental data[Section 5,[ACNL⁺23]] is available for $l = 2, 3$. The related sets to study l^i isogenies ($i \geq 0$) between conjugates is S_i [Equation 5]. The cardinality of such sets [Theorem 23, Theorem 24] have been well studied for $i = 0, 1$. We see that studying such sets for $i > 1$ as on increase of cardinality of the target set minimum length of pseudo-random walk, used generally in path-finding problem will decrease [Proposition 22], and we have already proved $i = 3$ is the optimal value.

- We have investigated [Tables 2, 3 and 4] the paths between frobenius conjugates for $l > 2, 3$. We introduce sets $\overline{S}_i$ [Equation 6], the generalised form of the sets S_i , which adapts to the presence of short cycles. Our studies partly fulfill the requirement specified in Section 4.1 of [ACNL⁺23], which states that “Further studies on a broader sample of primes would be required to fully explore the differences between the distributions of distances between conjugate and arbitrary vertices.”

Some recent works [ABC⁺25, KKA⁺24] deal with cycle computation using several prime degrees of isogenies. The bottleneck still lies in the choice of primes that would be necessary to generate cycles for constructing endomorphism rings of the general case of curves, *i.e.*, supersingular elliptic curves over $\mathbb{F}_{p^2}$. We stick to one prime l and the necessary pre-computations that would minimize the time.

1.2 Paper organization

We introduce the context of the problem in Section 1. We specifically point out our contributions in subsection 1.1. In Section 2, we give some background regarding elliptic curves and isogenies. In Section 3, we present the specific problem of path finding to conjugates and the motivation to work with it. In A, we analyse the relative cardinalities of some sets which arise when short cycles are present. We calculate the cost of finding prime power isogenies in Section 4, where we also devise the optimal way in which we can do the test for finding path to conjugate. We outline the importance of choosing l beforehand and the pre-computation strategies to do so in Section 6. We list some future works in Section 7.

2 Preliminaries

In this section, we introduce the necessary results needed for the rest of the paper. A detailed background on elliptic curves, isogenies and related information is available in literature [Sil86, Was08, Gal12, DF17, Sot20, Mey21, Pan21]. Throughout the paper, we shall denote the algebraic closure of any field K as $\overline{K}$. We state our results over the finite field $\mathbb{F}_p$ or over it’s algebraic closure, where in general p is a large prime. Most of them can be generalized for any arbitrary field, with some more modification.

2.1 Elliptic curve

Elliptic curves are projective curves of genus 1 with a specific base point. The advantage of using projective space is to better understand the process of adding *points at infinity*. It helps to realize the

geometry of projections. Less formally, we can think of projective space as the collection of all lines in the affine space passing through the origin.

Definition 1. An elliptic curve E over a finite field $\mathbb{F}_{p^n}$ ($n \in \mathbb{N}$) is given by a (short) Weierstrass form

$$E(\mathbb{F}_{p^n}) := \{(x, y) \in \mathbb{F}_{p^n}^2 : y^2 = x^3 + ax + b \text{ where } a, b \in \mathbb{F}_{p^n}\} \cup \{\mathbf{0}_E\} \quad (1)$$

where $\mathbf{0}_E$ is the point $(x : y : z) = (0 : 1 : 0)$ on the projective curve $y^2z = x^3 + Axz^2 + Bz^3$.

When the curve is fixed, we will denote $\mathbf{0}_E$ as the zero of the curve. Also, when we say that the elliptic curve is considered over the algebraic closure $\overline{\mathbb{F}_{p^n}}$, we shall consider the points $(x, y) \in \overline{\mathbb{F}_{p^n}}$.

We see that the set of points in an elliptic curve are solutions to a degree three equation. From Bézout's theorem, any line in the two dimensional projective space will intersect the curve in exactly three points, taken with multiplicity. This gives a geometric way to define a group structure on an elliptic curve.

Definition 2. Let $E(\mathbb{F}_{p^n}) : y^2 = x^3 + ax + b$ where $a, b \in \mathbb{F}_{p^n}$ be an elliptic curve. Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be two points on E , which are not $\mathbf{0}_E$. We can define a composition law $\oplus$ as follows:

- $P \oplus \mathbf{0}_E = \mathbf{0}_E \oplus P = P$ for any point $P \in E$;
- If $x_1 = x_2$ and $y_1 = -y_2$ then $P_1 \oplus P_2 = \mathbf{0}_E$;
- If the above does not hold true, let $\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & \text{if } P_1 \neq P_2 \\ \frac{3x_1^2 + a}{2y_1} & \text{if } P_1 = P_2 \end{cases}$
and set $(P_1 \oplus P_2) = (x_3, y_3)$ where

$$\begin{aligned} x_3 &= \lambda^2 - x_1 - x_2 \\ y_3 &= -\lambda x_3 - y_1 + \lambda x_1. \end{aligned}$$

Using the above definition, the operation $\oplus$ induces a group structure on E . Let us summarize the above as

Proposition 1. [Chapter III, Proposition 2.2, [Sil09]] The set of points on E form an abelian group under the operation $\oplus$, with identity element $\mathbf{0}_E$. The n -th scalar multiple point of any point P , is denoted by $[n]P = (P + P + \dots P(n \text{ times}))$ and the inverse of the point $P = (x, y)$ is the point $-P = (x, -y)$.

Definition 3. Given an elliptic curve and a point $P \in E$, the order of the point P is the smallest positive integer k such that $[k]P = \mathbf{0}_E$.

Definition 4. Torsion subgroup: The d -torsion subgroup, denoted by $E[d]$ is the set of points having d -th scalar multiple point as the zero element, i.e.,

$$E[d] = \{P \in E(\overline{\mathbb{F}_p}) \mid [d]P = \mathbf{0}_E\}$$

Theorem 2 (Chapter III, Corollary 6.4, [Sil09]). The structure of the torsion group can be organized as:

- If $\gcd(p, d) = 1$, $E[d] \cong \mathbb{Z}/d\mathbb{Z} \times \mathbb{Z}/d\mathbb{Z}$.
- For supersingular elliptic curves E , $E[p^e] = \{\mathbf{0}_E\}$ for integers $e \geq 1$.

2.2 Maps between elliptic curves

We focus on maps between elliptic curves that preserves the structure both as projective varieties and groups. The invertible algebraic maps makes linear changes of coordinates that preserve the Weierstrass form of the equation and the group law. Such maps take the form $(x, y) \mapsto (u^2x', u^3y')$ for some element u in algebraic closure, and these maps define an *isomorphism* between the curves $y^2 = x^3 + au^4x + bu^6$ and $(y')^2 = (x')^3 + ax' + b$. A good way to represent such isomorphism classes is by using invariants.

Definition 5. Discriminant and j -invariant: The discriminant and j -invariant of an elliptic curve E is defined as $\Delta(E) = -16(4a^3 + 27b^2) \neq 0$ and $j(E) = -1728 \frac{(4a)^3}{\Delta(E)}$ respectively.

The definition of discriminant makes the j -invariant well-defined. Based on the above definitions, we shall explore certain properties, which we present as theorems below.

Theorem 3. For any $j_0 \in \mathbb{F}_{p^n}, \exists E_0$ over $\mathbb{F}_{p^n}$ such that $j(E_0) = j_0$.¹

Proof. Let us consider any $j_0 \in \mathbb{F}_{p^n}$ such that $j_0 \neq 0, 1728$.

It is easy to verify that the elliptic curve

$$E := y^2 = x^3 + \frac{3j}{1728 - j_0}x + \frac{2j_0}{1728 - j_0}$$

has $j(E) = j_0$.

The elliptic curve $E_0 : y^2 = x^3 + b$ has j -invariant 0, and the elliptic curve $E_{1728} : y^2 = x^3 + ax$ has j -invariant 1728. ■

When we work with *isomorphism classes* of elliptic curves, the above definition will help us to find a unique representative for each class.

Theorem 4. Two elliptic curves are isomorphic over $\overline{\mathbb{F}_{p^n}}$ if and only if they have the same j -invariant.

Proof. Isomorphisms between elliptic curves are of the form

$$(x, y) \mapsto (u^2x', u^3y') \tag{2}$$

Let us consider two isomorphic elliptic curves E_1, E_2 where

$$E_i := y^2 = x^3 + a_i x + b_i \text{ for } i = 1, 2 \text{ and } a_2 = u^4 a_1 \text{ and } b_2 = u^6 b_1.$$

It is easy to see that $j(E_1) = j(E_2)$. ■

We note that the above isomorphism holds over the algebraic closure. The cardinality of an elliptic curve is roughly equal to p^n . More specifically,

Theorem 5. [Hasse] The cardinality of E is $\#E(\mathbb{F}_{p^n}) = p^n + 1 - t$ where t is the trace of the p^n -th frobenius map. Also, $|t| \leq 2\sqrt{p^n}$.

Proof. Please refer Chapter V of [Sil09]. ■

¹Our prime p is a large prime. The proof here, however can be adapted for $p = 2, 3$ also.

The above theorem implies that the number of points on an elliptic curve over $\mathbb{F}_{p^n}$ lies in the interval $[p^n + 1 - 2\sqrt{p^n}, p^n + 1 + 2\sqrt{p^n}]$. This interval is called the Hasse interval. A further classification of the cardinalities for various values of extension degree when the elliptic curve is defined over an extension of the finite field $\mathbb{F}_p$, is there in [Wat69].

Definition 6. Let E be an elliptic curve with $p^n + 1 - t$ points, where t is the trace of the p^n -th Frobenius map.

- E is supersingular if $p|t$.
- Otherwise, E is called ordinary.

The above definitions also hold in terms of Weierstrass equation, torsion points, multiplication map, endomorphism rings or zeta function. Throughout this paper, our focus is on supersingular elliptic curves. We shall next see, that any such supersingular elliptic curves can be given in terms of elements of $\mathbb{F}_{p^2}$.

Theorem 6. Every supersingular elliptic curve over $\overline{\mathbb{F}_p}$ has its j -invariant defined over $\mathbb{F}_{p^2}$.

Proof. Please refer Theorem 3.1, Chapter V of [Sil09]. ■

Henceforth, we will assume the curves to be defined over $\mathbb{F}_{p^2}$. There are approximately $\frac{p}{12}$ isomorphism classes of supersingular elliptic curves defined over $\mathbb{F}_{p^2}$.

Definition 7. Isogeny: An isogeny $\phi : E_1 \rightarrow E_2$ between two elliptic curves E_1, E_2 is a non-constant rational map which is also a group homomorphism.

An example of isogeny is the multiplication by some $m \in \mathbb{Z}$, denoted by $[m]$ which takes any point P to mP obtained by m times adding P . The zero isogeny is the constant map which takes each point of E to $\mathbf{0}_E$. The kernel of an isogeny ϕ is

$$\text{Ker}(\phi) = \{P \in E | \phi(P) = \mathbf{0}_E\}$$

Theorem 7. [Chapter III, [Sil09]] Every non-zero isogeny $\phi : E_1 \rightarrow E_2$ between two elliptic curves E_1, E_2 , induces an injection of function fields $\phi^* : \overline{\mathbb{F}_{p^2}}(E_2) \rightarrow \overline{\mathbb{F}_{p^2}}(E_1)$.

The degree of an isogeny is defined from this injection, which leads to an extension field.

Definition 8. Degree: The degree of a non-zero isogeny is defined as $\deg \phi = [\overline{\mathbb{F}_{p^2}}(E_1) : \phi^* \overline{\mathbb{F}_{p^2}}(E_2)]$.

Theorem 8. [Theorem 9.6.3, [Gal12]] Any non-zero isogeny is surjective over the algebraic closure.

Theorem 9. [Lemma 9.6.4, [Gal12]] If ϕ is a non-zero separable isogeny of degree d , then it has a finite kernel of cardinality d .

By the definition of isogeny, which is a group homomorphism, this kernel is a subgroup of $E(\overline{\mathbb{F}_{p^2}})$. Definition 7 implies that any isogeny can be expressed in terms of rational functions $(\frac{p_1(x)}{q_1(x)}, \frac{p_2(x)}{q_2(x)} \cdot y)$ where $p_1, q_1, p_2, q_2 \in \mathbb{F}_{p^n}[x]$ with $\gcd(p_1, q_1) = 1$. A generalization of this statement holds true for any elliptic curve and the proof is available in Lemma 9.6.12 of [Gal12].

The *degree* of a non-zero isogeny is the degree of the morphism. The degree of the zero isogeny is zero. If there is an isogeny of degree d between curves E_1 and E_2 , then they are said to be *d-isogenous* or simply isogenous. Furthermore, a non-zero isogeny is separable if it is *separable* as a morphism [Definition 8.1.6, [Gal12]].

We can give a better representation of degree and separability when the isogeny is expressed in terms of rational functions as:

Lemma 10. [Lemma 9.6.13 [Gal12]] *The degree of an isogeny is given as $\deg(\phi) = \max(\deg p_1, \deg q_1)$ and ϕ is separable if $\left(\frac{p_1(x)}{q_1(x)}\right)' \neq 0$.*

Proposition 11 (Proposition 4.12, Chapter III, [Sil09]). *All isogenies of some prime degree $l \neq p$ are separable, and can be described by it's kernel.*

Isogenies with equal kernel are identified as same. We say that they are equivalent if one isogeny can be obtained from another by composing an automorphism. We can compute the isogeny from kernel by using Velu's formula [Vél71]. The complexity of finding a degree n isogeny using this formula is $O(n)$ field operations, and a more recent one being $O(\sqrt{n})$ [Ber20], which in certain applications for large degrees is exponential.

A nice way to compute isogenies, instead of using Velu's formula is to use the fact that isogenies can be written as a *chain* of prime degree isogenies. The specific way is as follows:

Theorem 12 (Theorem 25.1.2, [Gal12]). *Given a field K , two elliptic curve E_1, E_2 defined over K , and a separable isogeny $\phi : E_1 \rightarrow E_2$ defined over K , ϕ can be expressed as $\phi = \phi_1 \circ \phi_2 \circ \dots \phi_k \circ [n]$, where $\phi_1, \phi_2, \dots, \phi_k$ are isogenies of prime degree are defined over K and $\deg(\phi) = n^2 \prod_{i=1}^k \deg(\phi_i)$.*

This means that for an isogeny, written as the composition of k such isogenies, the cost to compute it is proportional to k , whereas the corresponding cost by using Velu's formula is $O(2^k)$. The general technique is to split each isogeny into product of smaller degree isogenies and to compute smaller degree isogenies with the help of Velu's formula.

Definition 9. Dual: *Given any separable isogeny $\phi : E_1 \rightarrow E_2$, there is an isogeny $\hat{\phi} : E_2 \rightarrow E_1$ such that $\phi \circ \hat{\phi} = \hat{\phi} \circ \phi = [\deg(\phi)]$. This map $\hat{\phi}$ is called the dual of ϕ .*

The property of being isogeneous induces an *equivalence* relation on the set of elliptic curves.

Definition 10. Endomorphism: *An isogeny from any elliptic curve E to itself is called an endomorphism of E .*

The multiplication-by- m map $[m]$, is an endomorphism for any integer m . We shall define now define an example of a non-scalar endomorphism which is the Frobenius endomorphism [Chapter III.4, [Sil09]].

Definition 11. Frobenius endomorphism: *For q , a power of the prime p , the q -power Frobenius endomorphism is the rational map*

$$\begin{aligned} \pi_q : E(\mathbb{F}_q) &\rightarrow E(\mathbb{F}_q) \\ \pi_q(\mathbf{0}_E) &= \mathbf{0}_E \text{ and } \pi_q(x, y) = (x^q, y^q) \end{aligned}$$

The frobenius endomorphism has some important properties: For points $P \in E(\mathbb{F}_q)$, the action of π_q is trivial, i.e., $\pi_q(P) = P$, whereas for points over any extension field, the action is non-trivial. A further generalization of this is that, for $P \in E(\overline{\mathbb{F}_q}) \setminus E(\mathbb{F}_{q^i})$, the action of π_q is non-trivial in the sense that $\pi_q^i(x, y) = (x^{q^i}, y^{q^i})$.

From above,

$$\begin{aligned} ([1] - \pi_q)P = P - \pi_q(P) = \infty &\iff P \in E(\mathbb{F}_q) \\ \implies \#E(\mathbb{F}_q) &= \#Ker([1] - \pi_q) \end{aligned}$$

The isogeny classes can be linked to the cardinality as below:

Theorem 13. [Sato-Tate] Two elliptic curves $E_1(\mathbb{F}_q)$ and $E_2(\mathbb{F}_q)$ are isogenous for a prime power q , if and only if $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

Theorem 14. [Chapter III.4, [Sil09]] The set of endomorphisms defined over $\overline{\mathbb{F}_p}$ together with the zero map, form a ring under the operations of addition and multiplication, and is called the endomorphism ring of E , which is usually denoted by $End(E)$.

$$End(E) = \{\phi : E \rightarrow E\} \cup 0$$

The addition of isogenies in this ring is defined by using elliptic curve additions as

$$(\phi_1 + \phi_2)(P) = \phi_1(P) + \phi_2(P)$$

The multiplication here is the usual composition operation.

For any elliptic curve, there is a neat description of its structure as below:

Theorem 15. [Corollary III.9.4, [Sil09]] For any elliptic curve E , the endomorphism ring $End(E)$ is either $\mathbb{Z}$ or order in an imaginary quadratic field or an order in a quaternion algebra².

Elliptic curves can be distinguished [Definition 9.11.3, [Gal12]] as supersingular or ordinary in terms of their endomorphism rings. Our focus in particular is on endomorphism rings of supersingular elliptic curves, which have a particular characterization as below:

Theorem 16. [Theorem 10.6, [Was08]] For a supersingular elliptic curve E , the endomorphism ring $End(E)$ is isomorphic to a maximal order in a quaternion algebra that is ramified at p and ∞ and split at the other primes.

Supersingularity of an elliptic curve can also be linked with cardinality and torsion points as below:

Theorem 17 (Theorem 9.11.2 [Gal12]). Let $E(\mathbb{F}_q)$ be an elliptic curve where q is a power of the prime p . Then the following conditions are equivalent

1. The endomorphism ring $End(E)$ is not commutative.
2. The p -th torsion point set $E[p] = \{\infty\}$.
3. $\#E(\mathbb{F}_q) = q + 1 - t$, where $p|t$.

The last condition in particular implies that the curve is supersingular and vice-versa.

²Please refer to [Voi21] for quaternion algebra's and related concepts.

2.3 Relations between computational problems

In this section we consider the problems of computing isogenies and endomorphism rings.

Definition 12. Isogeny computation problem: *Given the j -invariants $j(E)$ and $j(E')$ of elliptic curves E and E' over $\mathbb{F}_{p^2}$, find the isogeny $\phi : E \rightarrow E'$.*

The decisional version of the isogeny finding problem in Definition 12 is solvable in polynomial time by Theorem 13. When the above problem is solved and the output is a chain of l -isogenies of length $O(\log p)$, the endomorphism ring can also be computed [Section 6, [EHL⁺20]]. The converse is also true, *i.e.*, paths between two elliptic curves can be found if the endomorphism rings are known.

To conclude, the problems of endomorphism ring computation and isogeny computation are equivalent [EHL⁺18, Wes22] under polynomial time reductions. This means that given elliptic curves E , E' , and the endomorphism ring $\text{End}(E)$,

- If $\text{End}(E')$ can be known, then the isogeny between E and E' can be computed in polynomial time.
- If the isogeny between E and E' can be known, then $\text{End}(E')$ can be retrieved in polynomial time.

2.4 Modular polynomials

We have seen that isogenies can be computed using Velu's formula [Vél71] or by defining as a chain of isogenies in a convenient way. There is yet another way of finding isogenies, by using some special form of a bivariate polynomial called the *modular polynomial* that does not deal with kernel subgroups or points of elliptic curves.

The *modular polynomial* $\phi_N(x, y) \in \mathbb{Z}[x, y]$ for any integer $N \geq 2$ is a symmetric polynomial of degree $(N + 1)$, and its roots correspond to pair of N -isogeneous j -invariants [Exercise 2.18, Chapter II, [Sil94]]. For a prime N , it has degree $(N + 1)$, and is of the form

$$\phi_N(x, y) = x^{N+1} - x^N y^N + y^{N+1} + \sum_{k, m \leq N, k+m < 2N} a_{km} x^k y^m \text{ for } a_{km} \in \mathbb{Z}.$$

Given any positive integer $N = \prod_{i=1}^n l_i^{e_i}$,

$$\deg(\phi_N(x, y)) = \prod_{i=1}^n (l_i + 1) l_i^{e_i - 1}$$

The size of this polynomial is $O(N^3 \log N)$ bits, where the critical point is storing the coefficients, as they are generally taken modulo the prime p , which has a large size, due to its cryptographic application. For working purpose, we consider $\phi_N(x, y) \in \mathbb{F}_p[x, y]$. As these polynomials are symmetric, it is enough to store about half the coefficients, *i.e.*, around $\frac{N^2}{2}$ elements of $\mathbb{F}_p$, which needs $O(N^2 \log p)$ bits.

Given an elliptic curve E with j invariant j_1 , to compute some N -isogeneous curve E'_1 with j -invariant j'_1 , we note that by property of modular polynomials:

There will exist a separable isogeny $\phi : E_1 \rightarrow E'_1$ of degree N if and only if $\phi_N(j_1, j'_1) = 0$.

In our set-up, the j -invariants $j_1, j'_1 \in \mathbb{F}_{p^2}$. Also, by the existence of dual isogenies, $\phi_N(j'_1, j_1) = 0$.

Thus, a way to find isogeneous curves is to evaluate $\phi_N(j(E), x)$ and then compute roots. These polynomials can have large degree and coefficients, some such examples are available in [Sutherland's database](#). An algorithm due to Elkies [E+98] allows to compute the isogeny in time exponential in N . The modular polynomials can be connected to torsion subgroups as: There are $(l + 1)$ subgroups of the l -torsion group of an elliptic curve E , with j -invariant j . Each subgroup corresponds to isogeneous curve E' of degree l . The j -invariant of E' is the zero of the polynomial $\phi_l(j, x)$.

2.5 Isogeny graphs

We have informally introduced the concept of supersingular isogeny graph $\chi_l(\overline{\mathbb{F}_p})$ in Section 1. The $\mathbb{F}_{p^2}$ -isogeny-class or simply the isogeny class of an elliptic curve E , is the set of $\mathbb{F}_{p^2}$ -isomorphism classes of elliptic curves over $\mathbb{F}_{p^2}$ that are isogenous over $\mathbb{F}_{p^2}$ to E . By (Sato-Tate)'s Theorem (Theorem 13), this happens when the cardinalities are equal. The isomorphism classes can be represented by means of a graph.

Definition 13. Isogeny graph: *Let p be the prime characteristic over which the elliptic curves are defined and let $l \neq p$ be another prime. The l -isogeny graph $\chi_l(\overline{\mathbb{F}_p})$ is a directed multi-graph where the vertex set is the set of $\mathbb{F}_{p^2}$ -isogeny classes and the edges are the corresponding isogenies.*

The number of edges from any j -invariant is the $(l + 1)$ of roots of the modular polynomial evaluated at that point. The graph can be a multi-graph and for each edge $(j(E_1), j(E_2))$, the concept of dual isogeny implies that $(j(E_2), j(E_1))$ is also an edge, pointing to the opposite direction. When $j(E_1), j(E_2) \neq 0, 1728$, the multiplicities of $(j(E_1), j(E_2))$ and $(j(E_2), j(E_1))$ are same, meaning that barring $\{0, 1728\}$ we can view it as an undirected graph by identifying isogenies with it's dual and edges as equivalent upto post composition with an automorphism. The number of edges differ when the j invariants 0, 1728 are present, due to the presence of extra automorphisms [Remark 25.3.2 [Gal12]].

The isogeny graph has a nice structure which we summarize in the proposition below:

Proposition 18. *The graph $\chi_l(\overline{\mathbb{F}_p})$ is connected [Section 25.3.3, [Gal12]], expander graph and has Ramanujan property [Piz90].*

The above proposition means that the graph has good mixing properties, which implies that the probability of any random walk, of size same as the diameter of the graph, to reach any arbitrary vertex is close to uniform. More specifically, this connected graph is $(l + 1)$ -regular(except at 0, 1728 due to extra automorphisms).

Any two elliptic curves in this graph is connected by chain of m isogenies of degree l where $m = O(\log p)$ [Theorem 79, [Koh96]]. For a fixed prime $l (= O(\log p))$, any l -isogeny in the just mentioned chain can be specified by rational maps or kernel, and so have representations of size polynomial in $\log p$.

By Theorem 6, there can only be only finite number of isomorphism classes of graph, which means that the size of vertices is finite. The exact number can be obtained from the following proposition.

Proposition 19. *[Corollary 4.40, [Was08]] The cardinality of the set V of vertices of this graph is $\#V = \lfloor \frac{p}{12} \rfloor + \epsilon_p$ where $\epsilon_p = 0, 1, 2$ accordingly as $p \equiv 1, \{3, 5, 7\}, 11 \pmod{12}$ respectively.*

This graph is a Ramanujan graph [Piz90], an optimal expander graph. The mixing of random walks is widely used in endomorphism ring computation and path-finding problems.

2.5.1 Isogeny cycles

Before, we define cycles, we need the concept of backtracking. We have already defined dual isogenies in Definition 9. Two edges are called dual if the dual of the isogeny corresponding to it is equal to the composition of automorphism of it with the isogeny corresponding to the other. By *backtracking* we mean an edge is dual to any other.

Definition 14. Isogeny cycle: *An isogeny cycle is a closed walk containing the given set of vertices of elliptic curves, which contains no backtracking and is not a power of another closed walk.*

From now onwards, which we shall refer an isogeny cycle as simply cycle. The cycle $\{a_1, a_2, a_3, \dots, a_N\}$, through some elliptic curve E , is therefore an isogeny from E to E and so corresponds to an endomorphism in $\text{End}(E)$ of degree l^N .

2.5.2 Conjugates

We have defined the isogeny π_q in Definition 11. We are now in a position to define frobenius conjugate of an elliptic curve.

Definition 15. Conjugate: *Given an elliptic curve $E : y^2 = x^3 + ax + b$ over $\mathbb{F}_{p^2}$, the frobenius conjugate or conjugate is another elliptic curve given by $E^{(p)} : y^2 = x^3 + a^p x + b^p$.*

The Frobenius morphism $\text{Frob} : E \rightarrow E^{(p)}$ is given by $(x, y) \mapsto (x^p, y^p)$. The Galois group $\text{Gal}(\mathbb{F}_{p^2}/\mathbb{F}_p)$ of the frobenius automorphism $\pi : \mathbb{F}_{p^2} \rightarrow \mathbb{F}_{p^2}$ such that $\pi(\alpha) = \alpha^p$ acts on the graph $\chi_l(\overline{\mathbb{F}_p})$ and sends a j -invariant j_1 to j_1^p (the conjugate). The fixed points under this automorphism are the supersingular j -invariants in $\mathbb{F}_p$. Any isogeny $\Phi : E_1 \rightarrow E_2$ results in another isogeny $\Phi^{(p)} : E_1^{(p)} \rightarrow E_2^{(p)}$ between the conjugate curves. Also, $(\Phi^{(p)})^p = \Phi$. The mirror involution [Definition 2.4, [ACNL⁺23]] sends each element of the graph $\chi_l(\overline{\mathbb{F}_p})$ to its conjugate, resulting in some important structural properties. The frobenius automorphism π , is an involution.

Lemma 20. [Lemma 6, [CLG09]] *The algebraic number theoretical condition of E and $E^{(p)}$ being l -isogeneous for some supersingular elliptic curve E over $\mathbb{F}_{p^2}$, is that $\mathbb{Z}[\sqrt{-lp}]$ embeds into $\text{End}(E)$.*

2.6 Isogenies over subfields

Considering the isogeny computation problem in Definition 12, when the elliptic curves are considered over the subfield $\mathbb{F}_p$, instead of $\mathbb{F}_{p^2}$, the time requirement is $O(p^{\frac{1}{4}})$ [DG16]. This algorithm can be used to solve the general supersingular isogeny computation problem of Definition 12 in time $O(p^{\frac{1}{2}})$. Another version of the problem is to compute isogenies between supersingular elliptic curves E over the field $\mathbb{F}_{p^2}$ and E' over the subfield $\mathbb{F}_p$. This is essentially a sub-routine of the general isogeny computation problem. The Supersolver algorithm of [CRSCS22] uses modular polynomials $\phi_l(x, y)$, for various values of prime l . The main strategy is to test roots in subfield without finding them explicitly at first. This reduces the time of general isogeny computation practically when compared to using a single modular polynomial for exactly one prime l for finding isogenies at each step.

3 Path finding to conjugates in supersingular isogeny graphs

We shall now discuss mirror paths and certain sets associated with path finding to conjugates. Next, we introduce a modified version of these sets, which are useful from the cryptographic security viewpoint.

3.1 Mirror paths

The definition of modular polynomials implies that

$$\phi_l(j_1, j_2) = 0 \implies \phi_l(j_1^p, j_2^p) = \phi_l(j_1, j_2)^p = 0 \quad (3)$$

Recursively, using the above equation, cycles are build with one side the mirror image under the frobenius map, of the other side

Definition 16. Mirror path: [Definition 2.6 [ACNL⁺23]] An undirected path $\{j_0, j_1, \dots, j_n\}$ is a mirror path if it is invariant under the mirror involution.

This now implies that for l -isogeneous vertices j_1 and j_2 , if any of them is in $\mathbb{F}_p$ or isogeneous to it's conjugate, then there exists a path between the other vertex and it's conjugate. When this event happens in a pseudo-random walk, then, the path can be traced backwards to reach the starting vertex. Mirror paths are either of the form $j_1 \rightarrow j_2 \rightarrow \dots j_m \rightarrow j \rightarrow j_m^p \rightarrow j_{m-1}^p \rightarrow j_2^p \rightarrow j_1^p$ (where the intermediate vertex j is in $\mathbb{F}_p$), which makes it of even length or it is of the form $j_1 \rightarrow j_2 \rightarrow \dots j \rightarrow j^p \rightarrow j_2^p \rightarrow j_1^p$ (where j is l -isogeneous to j^p) of odd length.

The above discussion also points the nature of length one paths. We formally write it in the lemma below.

Lemma 21. j, j^p is l -isogeneous means there is some root of the modular poly $\phi_l(j, x) \pmod p$ in the subfield

An essential subroutine of cycle finding [Algorithm 3.4, [EHL⁺20]] is to find a path between conjugate vertices in $\chi_l(\overline{\mathbb{F}_p})$ for fixed l . This makes it relevant to study values of l and minimum value of B (for fixed l), so that finding l^B isogeny to the conjugate j^p takes optimal time.

3.2 Length of pseudo-random walk

The length of the pseudo-random walk to find a cycle, can be connected to the cardinality of the target vertex set as below.

Proposition 22. [Proposition 3.6, [EHL⁺20]] Let us consider a random walk in $\chi_l(\overline{\mathbb{F}_p})$ ($p > 3$ and $l \neq p$) to reach a set S of vertices, not containing 0 or 1728. Then the length L of the random walk,

initiating from any arbitrary point has to be at least $L = \frac{\log\left(\frac{p-1}{6|S|^{\frac{1}{2}}}\right)}{\log\left(\frac{l+1}{2\sqrt{l}}\right)}$ to reach S with some probability $Pr_S(\geq \frac{6|S|}{p})$.

The above proposition implies that the cardinality of the set S is quite important. In the path-finding problem between conjugates, the related sets consists of curves with l^i ($i \geq 0$) isogeny to the conjugate curve.

3.2.1 Some cardinalities

Let us recall the studies made in literature regarding the cardinalities of such sets as in above section. From the proof of Lemma 20, the cardinalities of such sets is connected to the class number.

For $i = 0$, it is the set of all supersingular j -invariants in the subfield $\mathbb{F}_p$, where the curves are equal to their conjugates.

Theorem 23. [Equation 1, [DG16]] *The cardinality of the set S_0 of all supersingular j -invariants in $\mathbb{F}_p$, is:*

$$\#S_0 = \begin{cases} \frac{1}{2}h(-4p) & \text{if } p \equiv 1 \pmod{4}, \\ h(-p) & \text{if } p \equiv 7 \pmod{8}, \\ 2h(-p) & \text{if } p \equiv 3 \pmod{8}. \end{cases} \quad (4)$$

where $h(d)$ is the class number of the imaginary quadratic field $\mathbb{Q}(\sqrt{d})$.

The above class number is $O(p^{\frac{1}{2}})$ [Sie35], which implies that the size $\#S_0$ is $O(p^{\frac{1}{2}})$, where the actual value depends on the cardinality of the associated class group and the value of p modulo 8.

The above value is independent of the choice of the prime l , as it does not deal with any isogeneous curves. If we consider, l^i ($i \geq 1$) isogeny to the conjugate curve, we get sets S_i , where,

$$S_i = \{j \in \mathbb{F}_{p^2} : j \text{ is supersingular and is connected to its conjugate } j^p \text{ by an isogeny of degree } l^i\} \quad (5)$$

For example, [EHL⁺20] considers l -isogeny to the conjugate.

Theorem 24. [Theorem 3.9, [EHL⁺20]] *The lower bound on the cardinality of the corresponding set S_1 is $\#S_1 > \frac{C\sqrt{p}}{\log(\log(p))}$ where $l < \frac{p}{4}$ and $C > 0$ is a constant depending upon the value of l .*

In general, asymptotically, the cardinality [[CLG09], Lemma 6] of S_i is $l^{\frac{i}{2}}O(\sqrt{p})$.

3.3 Necessity to choose l and i optimally

A vital issue here is to test membership of an element in S_i . Clearly, this is related to the structure of the isogeny graph. Let us recall the previous result of [Koh96] in this regard.

Theorem 25. [Theorem 79, [Koh96]] *For every prime l , the graph $\chi_l(\overline{\mathbb{F}_p})$ has diameter $= O(\log p)$, where the constant in the bound is independent of l .*

This means for larger i ($= O(\log p)$), the existence of isogeny for any element of $\chi_l(\overline{\mathbb{F}_p})$ is guaranteed. But for smaller bounds on i , the isogeny of degree l^i between conjugates may not exist. The necessity of relatively larger l and i may arise as it increases the cardinality of S_i , which decreases the length of the pseudo-random walk. This cost of isogeny computation [Section 2.2], to test existence, grows bigger as the value of l or i increases. This motivates to study ways to choose l and i so that the total cost which depends both on the length of the pseudo-random walk and the cost of isogeny computation is optimally balanced, which has the possibility of adding practical improvements to the most optimized algorithm [EHL⁺20] for finding paths between conjugates.

3.4 Paths leading to mirror paths

The vertices in S_0 or S_1 give rise to mirror paths which aid in finding cycles. For $i > 1$, the paths corresponding to vertices in S_i may be mirror paths or non-mirror paths. Non-mirror paths may happen when a vertex is l -isogeneous to the conjugate of some vertex $\bar{j}$, whose immediate ancestor is a part of the pseudo-random walk, but $\bar{j}$ itself is not a part of the walk. Let us call such paths as *generalised sibling paths*.

Definition 17. Sibling Path: Suppose j is a vertex which appears in a pseudo-random walk with l -isogeneous supersingular j -invariants (leaving the one already chosen as it's ancestor) $j_{i_1}, j_{i_2}, \dots, j_{i_l}$. We call a path sibling path if $\exists i_1 \neq i_2$ such that $j_{i_1}^p$ and j_{i_2} are l -isogeneous.

In case of sibling paths j_{i_1} and j_{i_2} have a common ancestor at the immediate step. The notion of generalised sibling paths captures the cases when the immediate common ancestor may not be the same. Obviously, any sibling path is also a generalised sibling path.

For example, let, $j, j_{11}, j_{21}, j_{31}, j_{41}, j_{52}, j_{61}, j_{72}, j_{81}$ be a path such that j_{km} (obtained as a root of $j_{k-1,}$.) signifies a vertex at the k -th level and is the m -th of the l isogenous vertices. Following this notation, the vertex j_{32} is l -isogenous to j_{21} and it is not a part of the pseudo-random walk. Let the conjugate j_{32}^p be l isogeneous to j_{81} , i.e., $\phi_l(j_{81}, j_{32}^p) = 0$. We then obtain a short cycle³ as $j_{21}, j_{31}, j_{41}, j_{52}, j_{61}, j_{72}, j_{81}, j_{32}^p, j_{21}^p, j_{31}^p, j_{41}^p, j_{52}^p, j_{61}^p, j_{72}^p, j_{81}^p, j_{32}, j_{21}$. Thus we can observe that,

Remark 1. Any sibling path or generalised sibling path leads to a short cycle.

From above, paths between l^i conjugates for $i > 2$ can also be seen.

Lemma 26. For $i = 3$, the question is to know whether $\phi_l(j, x)$ and $\phi_l(x, x^p)$ have a common root.

Proof. Let α be the common root of $\phi_l(j, x)$ and $\phi_l(x, x^p)$.

Then, $j \rightarrow \alpha \rightarrow \alpha^p \rightarrow j^p$ is the path of length 3. ■

Let us recall the relation between isogeny and endomorphism ring.

Theorem 27. [Theorem 3.1 [MMP24]] Let the Generalised Riemann Hypothesis hold true. Also, let E, E' be two supersingular elliptic curves with an isogeny of degree l between E and E' . Furthermore, let $\text{End}(E)$ and $\text{End}(E')$ be the endomorphism rings of E and E' respectively.

If $\text{End}(E)$ and $\text{End}(E')$ are known, then the l -isogeny can be computed in polynomial time.

If the l -isogeny is known, along with one of the endomorphism rings, then the other endomorphism ring can be computed in polynomial time.

The above theorem implies that if a short cycle through any vertex of $\chi_l(\overline{\mathbb{F}_p})$ is known along with an isogeny from any other vertex, it may introduce vulnerabilities, which leads to knowledge about the endomorphism ring, which again aids in computing isogenies. This disturbs the foundational hardness assumptions of isogeny-based cryptography. Hence, we are interested in studying paths in $S_i (i > 1)$, which fall in the category of non-sibling paths. This implies the study of sets $S_i (i > 1)$, such that vertex $j \in S_i$ is only characterized by the fact that it has a descendant vertex in either of S_0 and S_1 . We exclude those vertices which may lead to generalised sibling paths. Since a new vertex is selected at every point of the pseudo-random walk, choosing such a vertex j , having immediate neighbour in S_0 or

³We express this as an isogeny with initial and terminal vertex j_{21}

S_1 earlier in the pseudo-random walk will save the cost of computing isogenies which may not lead to mirror paths or the paths which may be insecure for cryptographic purposes. Thus our focus will be to study the sets $\overline{S}_i = \{ j \in S_i (i > 1) \mid j \text{ has a descendant vertex in } S_0 \text{ or } S_i \}$. By the theory of mirror paths,

$$\overline{S}_i = \{ j \in S_i (i > 1) \mid j_{\frac{i}{2}} \in S_0 \text{ if } i \text{ is even and } j_{\lfloor \frac{i}{2} \rfloor} \in S_1 \text{ if } i \text{ is odd} \} \quad (6)$$

⁴ We thus discard elements of the set $S_i \setminus \overline{S}_i$ of those j -invariants which have descendants at any two levels (may be same also) such that one is isogeneous to the Frobenius conjugate of the other. Technically,

$$S_i \setminus \overline{S}_i = \{ j \in S_i (i > 1) \mid j \text{ has no descendant } j_{k_1, i_1} \text{ and } j_{k_2, i_2} \text{ such that } j_{k_1, i_1} \text{ is isogeneous to } j_{k_2, i_2}^p \} \quad (7)$$

⁵

4 Optimizing cost of finding an isogeny of degree l^i

The aim of this section is to find the cost of testing whether a given vertex j is l^i isogeneous to its conjugate j^p . The intuition is that, for a fixed l , we would like to reduce the length of the pseudo-random walk and accelerate the process of arriving at a suitable vertex that would give a path to the conjugate. We would like to minimize root computations, if possible. For smaller values of i , we consider the modular polynomial while for larger i , we use minimal polynomial generated from a system of modular polynomials. We calculate corresponding costs in terms of the multiplications needed⁶. There is a trade-off between increasing the value of i and testing whether elements are in $\overline{S}_i$. We compare the methods to identify time-wise efficient procedure.

Lemma 28. *Given an element in the finite field $\mathbb{F}_{p^2}$, the time complexity of testing whether it belongs to the subfield $\mathbb{F}_p$ or not, depends on the representation of the finite field.*

Proof. Let us consider various representations of $\mathbb{F}_{p^2}$.

Polynomial basis representation over $\mathbb{F}_p$: In this representation,

$$\mathbb{F}_{p^2} \cong \mathbb{F}_p[\alpha] / \langle f(\alpha) \rangle,$$

where $f(x)$ is an irreducible quadratic polynomial over $\mathbb{F}_p$.

Any element $j \in \mathbb{F}_{p^2}$, will then be expressed as

$$j = a + b\alpha \text{ where } a, b \in \mathbb{F}_p \text{ and } \alpha \text{ is a root of } f(x). \quad (8)$$

The action of the Frobenius map takes j to j^p .

$$j^p = (a + b\alpha)^p = \sum_{k=0}^p \binom{p}{k} a^{p-k} (b\alpha)^{(k)} \quad (9)$$

⁴where j_k represents any descendant at the k -th step

⁵where j_{t, i_1} represents a descendant at the t -th step

⁶We have neglected the costs of addition and subtraction in our analysis as they are quite less than those of multiplication.

The binomial coefficient $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ is divisible by p for all $1 \leq k \leq (p-1)$.

The above means, in $\mathbb{F}_p$, $\binom{p}{k} = 0$.

By Fermat's Little Theorem, $a^p = a$ and $b^p = b$.

Using these considerations in equation 9,

$$j^p = a^p + (b\alpha)^p = a + b\alpha^p \quad (10)$$

Considering the irreducible quadratic polynomial $f(x)$ as $f(x) = x^2 + a_1x + a_0$ with root α ,

$$(f(\alpha))^p = ((\alpha^2) + a_1\alpha + a_0)^p = (\alpha^2)^p + a_1^p\alpha^p + a_0^p \text{ [By equation 9]}$$

$$\implies (f(\alpha))^p = (\alpha^p)^2 + a_1(\alpha^p) + a_0 \text{ [By Fermat's Little Theorem]} = f(\alpha^p) \quad (11)$$

Now,

$$f(\alpha) = 0 \implies (f(\alpha))^p = 0$$

By equation 11, α^p is also a root of $f(x)$. Then, by applying Vieta's formula to the quadratic polynomial f ,

$$\alpha^p = -a_1 - \alpha \quad (12)$$

From equations 10 and 12,

$$j^p = a + b(-a_1 - \alpha) = (a - ba_1) + \alpha(-b)$$

Testing: We know that vertices in subfield $\mathbb{F}_p$ are equal to their conjugates, *i.e.*,

$$j \in \mathbb{F}_p \iff j^p = j \quad (13)$$

From Equations 8 and 13, this simplifies to testing whether,

$$\left(a \stackrel{?}{=} (a - ba_1) \text{ and } b \stackrel{?}{=} -b \right) \iff b \stackrel{?}{=} 0$$

The above comparison is clearly $O(1)$.

Normal basis representation In this case, $\mathbb{F}_{p^2}$ is represented using a basis of the form $\{\beta, \beta^p\}$ where $\beta \in \mathbb{F}_{p^2}$ has a minimal polynomial of degree two.

Any arbitrary $j \in \mathbb{F}_{p^2}$, is expressed as $j = a\beta + b\beta^p$, where $a, b \in \mathbb{F}_p$.

Equating elements in subfields with conjugates,

$$\begin{aligned} j = a\beta + b\beta^p &\implies j^p = b\beta + a\beta^p \\ j \in \mathbb{F}_p &\iff j = j^p \iff a = b \end{aligned}$$

This is again comparing two elements of $\mathbb{F}_p$, hence the cost is $O(1)$.

“Black Box” or Cryptographically Obfuscated Representation: In this case, the elements of $\mathbb{F}_{p^2}$, is represented in a encrypted or obfuscated way. Some general characteristics of such a representation is:

- The embedding of $\mathbb{F}_p$ into $\mathbb{F}_{p^2}$ is not known.
- The co-ordinates of any arbitrary points cannot be accessed.
- Fundamental field operations are: addition, multiplication and inverses.

Similar as before, testing any arbitrary $j \in \mathbb{F}_{p^2}$, is an element of $\mathbb{F}_p$ or not is seeing whether $j^p \stackrel{?}{=} j$. The disadvantage of obfuscation is that computing j^p will take $O(\log p)$ multiplications, except when p is extremely small or of some special form which permits some pre-computation techniques.

Representation via a multiplicative generator: Let g be a multiplicative generator of the multiplicative group $\mathbb{F}_{p^2}^*$.

Any non-zero $j \in \mathbb{F}_{p^2}$, is of the form $j = g^k$ for some integer $1 \leq k \leq (p^2 - 1)$. The subfield $\mathbb{F}_p^*$ is the unique cyclic sub-group of order $(p - 1)$.

$$j \in \mathbb{F}_p \iff j^{(p-1)} = 1 \iff g^{k(p-1)} = 1 \quad (14)$$

As the order of g is $(p^2 - 1)$,

$$(p^2 - 1) | k(p - 1)$$

Since $(p^2 - 1) = (p - 1)(p + 1)$, the above condition means $(p + 1)$ divides k . The cost of this is in between $O(\log p(\log(\log p)))$ and $O((\log p)^2)$.

Summing up the above discussion, the best way to test for subgroup membership is to use polynomial or normal basis representation, where it takes constant time. ■

Proposition 29. *Given a supersingular j -invariant j in $\mathbb{F}_{p^2}$, testing for a l^0 isogeny to its conjugate takes constant time.*

Proof. In order to test for a l^0 isogeny to conjugate, we require to test whether $j \in \mathbb{F}_p$ or not.

The proof follows from Lemma 28, by considering a polynomial or normal basis representation of $\mathbb{F}_{p^2}$. ■

Lemma 30. *Given a supersingular j -invariant j , the cost to find its conjugate is constant time.*

Proof. Let us represent $\mathbb{F}_{p^2} = \mathbb{F}_p(\beta)$, where β is a primitive element of the extension.

For more clarity, let us choose $\beta^2 = -1$ or the first non-square in the sequence $\pm n$, $n \geq 2$.

We can express the given j -invariant as $j = a + \beta b$ for $a, b \in \mathbb{F}_p$.

In the way β is chosen,

$$j = a + \beta b \implies j^p = a - \beta b$$

The operation on the right side costs $O(1)$. ■

Lemma 31. *Using the representation of the finite field $\mathbb{F}_{p^2}$ as in Lemma 30, the cost of multiplying two elements of $\mathbb{F}_{p^2}$ can be approximated by three multiplications in $\mathbb{F}_p$*

Proof. Let $u = x_0 + x_1\beta, v = y_0 + y_1\beta$ be two arbitrary elements of $\mathbb{F}_{p^2}$, where $x_0, x_1, y_0, y_1 \in \mathbb{F}_p$. We use the Karatsuba's method of multiplication,

$$\begin{aligned} u \cdot v &= (x_0 + x_1\beta) \cdot (y_0 + y_1\beta) \\ &= (x_0y_0 + x_1y_1) + (x_0y_1 + x_1y_0)\beta \end{aligned}$$

Let us denote the element obtained after multiplication as:

$$\begin{aligned} M_1 &= x_0y_0 \\ M_2 &= x_1y_1 \\ M_3 &= x_0y_1 + x_1y_0 \end{aligned}$$

The trick lies in the method of obtaining the third element from the first two. Clearly,

$$M_3 = (x_0 + x_1)(y_0 + y_1) - M_1 - M_2$$

Each of $M_i, i = 1, 2, 3$ thus requires one multiplication, which implies that the product of u, v can be done using 3 multiplications over $\mathbb{F}_p$. $\blacksquare$

Lemma 32. *The cost to evaluate the modular polynomial at conjugates is $O(l^2)$ multiplications in $\mathbb{F}_p$.*

Proof. The modular polynomial is of the form

$$\phi_l(x, y) \mod p = x^{l+1} - x^l y^l + y^{l+1} + \sum_{k, m \leq l, k+m < 2l} a_{km} x^k y^m \quad (15)$$

where the integers a_{km} are reduced modulo p .

Let j be the given supersingular j -invariant. In order to know, whether it is isogeneous to its conjugate $j^p (= \tilde{j}$, suppose), we need to evaluate $\phi_l(x, y) \mod p$ at $x = j, y = \tilde{j}$,

From equation 15, the above procedure will require the following steps:

- The power of j^k and $\tilde{j}^k$ for $2 \leq k \leq l$.
- Substituting the above powers and multiplying by the corresponding coefficients a_{km} .

For the first part, let us note that for $k > 1$, we can obtain the k -th power from the $(k-1)$ -th power and each such element belongs to $\mathbb{F}_{p^2}$.

The modular polynomial being symmetric, it is sufficient if we consider coefficients of powers of either x or y . For each $k \leq l$, the coefficient of x^k is of the form: $a_0 + a_1 y^1 + \dots + a_l y^l$, evaluating which costs $O(l)$ multiplications in $\mathbb{F}_{p^2}$. Considering all the l coefficients of x^k and the cost in Lemma 31, the total cost is $O(l^2)$ multiplications over $\mathbb{F}_p$. $\blacksquare$

Proposition 33. *Given a supersingular j -invariant j in $\mathbb{F}_{p^2}$, testing for a l -isogeny to its conjugate takes $O(l^2)$ multiplications over $\mathbb{F}_p$.*

Proof. To find whether l -isogeny exists between j and j^p , we need to know the value of $\phi_l(x, y) \mod p$ at $x = j, y = j^p$, and see whether it is zero.

By Lemma 32, knowing $v = \phi_l(j, j^p) \mod p$ is $O(l^2)$ multiplications over $\mathbb{F}_p$.

The comparison

$$v \stackrel{?}{=} 0$$

is $O(1)$. Thus the dominant cost to find isogeny between conjugates is $O(l^2)$ multiplications over $\mathbb{F}_p$. $\blacksquare$

Lemma 34. *Given a supersingular j -invariant j in $\mathbb{F}_{p^2}$, the cost of whether the modular polynomial $\phi_l(x, y) \bmod p$ at $x = j$ has a root in the subfield $\mathbb{F}_p$ is $O(l^2)$ multiplications in $\mathbb{F}_p$.*

Proof. To find root in $\mathbb{F}_p$, we first need to compute the univariate polynomial $f = \phi_{l,p}(j, x) \bmod p$. From the proof of Lemma 32, this cost is $O(l^2)$ multiplications in $\mathbb{F}_p$.

By Proposition 1 of [CRSCS22], we compute gcd of g_1, g_2 where $g_1 = f + \pi(f)$, $g_2 = f - \pi(f)$, and π is the p -power frobenius endomorphism.

π fixes each element of $\mathbb{F}_p$ and $\pi(\beta) + \beta = 0$ for the primitive element β , which implies $\pi^k(\beta) = (-1)^k \beta$. As multiplications by constants do not change the roots of the polynomials, we can modify the polynomials g_1, g_2 as:

$$\begin{aligned} g_1 &= \frac{1}{2}[\phi_l, p(j, x) + \pi(\phi_l, p(j, x))] \\ g_2 &= \frac{-\beta}{2}[\phi_l, p(j, x) - \pi(\phi_l, p(j, x))] \end{aligned}$$

Let $f = \sum_{i=0}^n a_i x^i$, where a_i 's are elements of $\mathbb{F}_{p^2}$, we see:

$$\begin{aligned} g_1 &= \frac{1}{2}[f + \pi(f)] = \text{Re}(a_n)x^n + \text{Re}(a_{n-1})x^{n-1} + \text{Re}(a_{n-2})x^{n-2} + \dots + \text{Re}(a_1)x^1 + \text{Re}(a_0) \\ g_2 &= \frac{-\beta}{2}[f - \pi(f)] = \text{Im}(a_n)x^n + \text{Im}(a_{n-1})x^{n-1} + \text{Im}(a_{n-2})x^{n-2} + \dots + \text{Im}(a_1)x^1 + \text{Im}(a_0) \end{aligned}$$

Using $(a + \beta b)^p = (a - \beta b)$, $\text{Re}(a + \beta b) = a$, $\text{Im}(a + \beta b) = b$ when each of the a_i 's are expressed in terms of β ,

$$g_1 = x^n + \dots + \text{Re}(a_0), \quad g_2 = \text{Im}(a_n)x^{n-1} + \dots + \text{Im}(a_0) \quad (16)$$

To test for root in $\mathbb{F}_p$, we compute the inverse-free gcd of g_1 and g_2 by using Algorithm 1 of [CRSCS22]. The time complexity is $O(l^2)$ [Proposition 2, [CRSCS22]] multiplications in $\mathbb{F}_p$. ■

Proposition 35. *Given a supersingular j -invariant j in $\mathbb{F}_{p^2}$, testing for a l^2 -isogeny to its conjugate takes $O(l^2)$ multiplications over $\mathbb{F}_p$.*

Proof. Let us recall Lemma 21. Then,

$$j \text{ is connected to } j^p \text{ by an } l^2 \text{ isogeny} \iff \phi_l(j, x) \bmod p \text{ has a root in the subfield } \mathbb{F}_p$$

Let $j_s \in \mathbb{F}_p$ be the j -invariant in the subfield. This is a root of both $\phi_l(j, x) \bmod p$ and $\phi_l(j^p, x) \bmod p$. From the proof of Lemma 32, the cost of computing $f = \phi_{l,p}(j, x)$ is $O(l^2)$ multiplications in $\mathbb{F}_p$.

By Lemma 34, the cost of testing for roots in subfield is $O(l^2)$ multiplications over $\mathbb{F}_p$. ■

Proposition 36. *Given a supersingular j -invariant j in $\mathbb{F}_{p^2}$, the only method of knowing whether it is connected to its conjugate, by l^i -isogeny for $i \geq 3$ is by computing roots.*

Proof. Let us consider $i = 3$ and j and j^p be connected by l^3 isogeny. By Lemma 26, the path can only be of the form $j \rightarrow \tilde{j} \rightarrow \tilde{j}^p \rightarrow j^p$ where $\tilde{j}$ is common root of both $\phi_l(j, x)$ and $\phi_l(x, x^p)$.

As $\deg(\phi_l(x, x^p)) = pl$, complexity to compute the greatest common divisor will be exponential in $\log p$. This implies that the only method that remains is to compute the root of the modular polynomial $\phi_l(j, x) \bmod p$.

For $i = 4$ and higher we need several modular polynomials, and the intermediate roots. ■

Lemma 37. *The cost of computing roots by standard root finding algorithms to find l -isogenies at each of d levels is $O(l^{(d+1)} \log p)$ operations in $\mathbb{F}_p$.*

Proof. Let us consider the following system of modular polynomials of size d :

$$\phi_l(j, j_1) = 0, \phi_l(j_1, j_2) = 0, \dots, \phi_l(j_{d-1}, j_d) = 0 \quad (17)$$

Let us exclude the root at the previous level and compute the possible roots at the next level. Let us compute roots till the $(d-1)$ -th level and obtain all supersingular elliptic curves at the d -th level.

This implies that for each modular polynomial, there are l options for j_1 , l^2 options for j_2 , and so on, with l^d options for j_d . Hence, total cost of root computation is $(1 + l + l^2 + \dots + l^{d-1})R_l$ where R_l is the cost of root computation for a degree l polynomial.

The standard algorithms for these are Berlekamp's algorithm [Ber70] with complexity $O(l^3 + l^2 \log l \log(p^2))$ operations in $\mathbb{F}_{p^2}$ or Cantor-Zassenhaus algorithm [CZ81] with complexity $O(l^3 \log(p^2))$ operations in $\mathbb{F}_{p^2}$. We can consider an optimized complexity of $O(l^3 + l^2 \log(p^2))$ in $\mathbb{F}_{p^2}$ [Sho09].

The deterministic variants are more costly. Replacing operations in $\mathbb{F}_{p^2}$ with those in $\mathbb{F}_p$, total complexity when we use the root-finding method is $\frac{1}{l-1}(l^d - 1)O(l^3 + l^2 \log p)$. The dominating factor in this case is $O(l^{(d+1)} \log p)$ operations in $\mathbb{F}_p$. ■

Lemma 38. *The complexity when using Gröbner basis algorithms is $O(d(l+1)^{3d})$ operations over $\mathbb{F}_p$ when solving the set of equations 17.*

Proof. The system of equations 17 will generate an ideal I in the multivariate polynomial ring where

$$I = \langle \phi_l(j, j_1), \phi_l(j_1, j_2), \dots, \phi_l(j_{d-1}, j_d) \rangle \subseteq \mathbb{F}_{p^2}[j_1, j_2, \dots, j_d]$$

As the modular polynomial for a prime $l \neq p$, can have at most $(l+1)$ distinct roots,

$$\dim(\mathbb{F}_{p^2}[j_1, j_2, \dots, j_d]/I) \leq (l+1)^d$$

By the finiteness theorem [Chapter 2, [CLO05]] and considering the facts that $j \in \mathbb{F}_{p^2}$ and I is zero-dimensional,

$$\exists \text{ non-zero polynomial in } I \cap \mathbb{F}_{p^2}[j_k] \ \forall \ k = 1, 2, \dots, d.$$

Let us now consider the minimal polynomial for the last variable with *lex* ordering $j_d > j_{d-1} > \dots > j_2 > j_1$. By Hilbert's basis theorem [Chapter 1, [CLO05]] and Buchberger's criteria [Chapter 1, [CLO05]], there exists a Gröbner basis.

By the FGLM algorithm [Chapter 2, [CLO05]], we can construct the Gröbner basis for the particular monomial ordering, and then find the minimal polynomial of j_d with respect to the ideal I .

Let m_d denote the minimal polynomial. We can then check whether the roots of m_d is in $\mathbb{F}_p$ or S_1 to ensure that a mirror path between j and j^p exists.

The complexity of using the FGLM algorithm is $O(d(l+1)^{3d})$ [Proposition 4.1, [FGLM93]] considering d variables and dimension $(l+1)^d$. ■

Lemma 39. *We can pre-compute the minimal polynomials and take advantage of existing generic relations and forego the cost in Lemma 38.*

Proof. We use the generic relation [Remark 3.1, [TKF⁺20]] that exists between modular polynomials and corresponding minimal polynomials⁷. For $d \geq 3$:

$$m_d(j, x) = \phi_{l^{d-2}}(j, x) \phi_{l^d}(j, x)$$

By using the above method, we can avoid computation of minimal polynomial at each step.

We pre-compute minimal polynomials m_d for some d from the knowledge of modular polynomials already pre-computed before⁸. This cost is one-time and removes the cost at each step.

Using same strategies as we evaluate modular polynomials and as $\deg(m_d) = O(l^d)$,

$$\deg(\phi_{l^{d-2}}(x, y)) + \deg(\phi_{l^d}(x, y)) = (l+1)l^{d-3} + (l+1)l^{d-1}.$$

Hence, the time to evaluate $m_d(x, y)$ at $x = j$ is $O(l^d)$ operations in $\mathbb{F}_p$. ■

The above method of using generic relations is cost-wise better than constructing the minimal polynomial at each step. We can test whether m_d has a root that gives rise to a mirror path between j and j^p .

The naive method would be to construct a system of $(i-1)$ equations like 17, and then test whether j^p is the root of this system by evaluating the corresponding minimal polynomial of degree $O(l^{i-1})$ at $x = j$ and $y = j^p$. The cost of this evaluation would be $O(l^{2(i-1)}) = O(l^{4d-2})$ where $i = 2d$ if it is even and $i = 2d+1$ if it is odd.

To reduce this complexity, we just concentrate on testing whether there are intermediate roots in $\mathbb{F}_p$ for even i or S_1 for odd i .

Lemma 40. *The cost of testing whether the roots of the minimal polynomial m_d is in $\mathbb{F}_p$ by using gcd [CRSCS22] computations without inverses is $O(l^{2d})$ operations over $\mathbb{F}_p$. Also, for $d \geq 2$, the above cost exceeds the cost to compute individual roots.*

Proof. Let us define g_1, g_2 as:

$$\begin{aligned} g_1 &= \frac{1}{2}[m_d(j, x) + \pi(m_d(j, x))] \\ &= \text{Re}(a_n)x^n + \text{Re}(a_{n-1})x^{n-1} + \text{Re}(a_{n-2})x^{n-2} + \dots + \text{Re}(a_1)x^1 + \text{Re}(a_0) \end{aligned}$$

$$\begin{aligned} g_2 &= \frac{-\beta}{2}[m_d(j, x) - \pi(m_d(j, x))] \\ &= \text{Im}(a_n)x^n + \text{Im}(a_{n-1})x^{n-1} + \text{Im}(a_{n-2})x^{n-2} + \dots + \text{Im}(a_1)x^1 + \text{Im}(a_0) \end{aligned}$$

for $m_d(j, x) = \sum_{i=0}^n a_i x^i$.

Let us apply the same analysis as in Lemma 34, by replacing modular polynomial by minimal polynomial. By using the relations between any element and its conjugate and adding the coefficients of β^0, β^1 ,

$$g_1(x), g_2(x) \in \mathbb{F}_p[x]$$

⁷We have run codes for some test cases, where we found this to hold true in all the cases.

⁸A database of modular polynomials: <https://math.mit.edu/~drew/ClassicalModPolys.html>

By applying Algorithm 1 of [CRSCS22] to g_1, g_2 , the cost of evaluating the gcd without inverse computation will be $O(l^{2d})$.

By Proposition 1 of [CRSCS22],

$$\deg(\gcd(g_1, g_2)) = 1 \implies \exists \text{ a root of } g_1, g_2 \in \mathbb{F}_p.$$

The range of l, d that we wish to consider is not too big, and so this is comparable to the analysis for the range of values of l considered in [CRSCS22]. This means the probability of the degree of the gcd being 1 is not negligible. The complete cost of evaluating a pre-computed minimal polynomial and knowing whether there is a root over $\mathbb{F}_p$ or not, is therefore,

$$O(l^d \log p) + O(l^{2d} \log p) = O(l^{2d} \log p)$$

For $d \geq 2$, the above cost is greater than the cost to compute individual roots. ■

Lemma 41. *The cost of testing whether roots of m_d is in S_1 is at least $O(l^{3d} \log p)$ operations.*

The maximum value of i for which we can test for isogenies between conjugates with cost $O(l^2 \log p)$ is $i = 3$.

Proof. To test whether roots of m_d is in S_1 , we test whether the polynomials $m_d(j, x)$ and $\phi_l(x, x^p)$ have common roots.

The roots that we are interested in are supersingular j -invariants over $\mathbb{F}_p^2$. We know that for any element $a + b\beta \in \mathbb{F}_{p^2}$ its p -th power is of the form $(a - \beta b)$. To remove p from the exponent, let us consider $M = m_d(j, a + \beta b)$ and $\Phi = \phi_l(a + \beta b, a - \beta b)$ as bivariate polynomials in variables a, b . Further, let

$$R = \text{Resultant}(M, \Phi, a)$$

Also, the degrees of the polynomial are:

$$\deg(M) = O(l^d), \deg(\Phi) = O(l)$$

Thus the construction of the Sylvester matrix will take about $O(l^{d+1})$ operations and the determinant will require about $O(l^{3d} \log p)$ operations.

Let us record the above steps in Algorithm 2. Using the same methods as in Lemma 37, total cost considering the cost of later steps of gcd and root computation exceeds the cost of individual root computation. This completes proof of the first part of the statement.

For $i = 3$ the value of $d(= \lfloor \frac{i}{2} \rfloor)$ is 1, so the cost in this case is $O(l^2 \log p)$. Beyond, this point, the cost increases. ■

Result 4.1. *The time complexity of testing l^i isogenies to conjugates is at most $O(l^2 \log p)$ for $i = 0, 1, 2, 3$, with complete root computation only required for $i = 3$. For $i > 3$ the cost is asymptotically higher.*

Proof. Suppose we are given a j -invariant j , and a positive integer i , and we have to test whether j and j^p are connected by l^i isogeny.

By Propositions 29, 33 and 35, the result holds true for $i = 0, 1, 2$.

The statements for $i \geq 3$ follows from Lemma 41. ■

Remark 2. *The above implies that the cost of testing whether vertices are in S_1, S_2, S_3 or not will take time complexity of $O(l^2 \log p)$.*

4.1 Optimal method to detect mirror paths: Best value of i and corresponding cardinality

The above proof indicates that the best value of i to balance the cost of computing isogenies with the advantage of viewing a longer path is $i = 3$. Above $i = 3$ the cost incurred exceeds $O(l^2 \log p)$ and grows further with an increase in i . Keeping the notations of equation 17, the set of our interest is $S_0 \cup S_1 \cup \{j \in S_2 | j_d \in S_0\} \cup \{j \in S_3 | j_d \in S_1\} = S_0 \cup S_1 \cup \overline{S_2} \cup \overline{S_3}$.

We have stated the entire procedure in Algorithm 1. We consider the non-trivial problem of Algorithm 3.4 of [EHL+20], *i.e.*, given a supersingular j_0 invariant in $\mathbb{F}_{p^2}$, we have to find a path to its conjugate j_0^p . Given a j -invariant we test first whether it belongs to $\mathbb{F}_p$ (Line 4). The cost of this is $O(1)$. If it is found, then the mirror path is already found. Else we now test whether it is in S_1 (Line 9) which will take time $O(l^2 \log p)$. Algorithm 3.4 of [EHL+20] considers upto this point. We shall now try to adapt the Supersolver technique [CRSCS22] in this gap.

4.1.1 Adaptation of the Supersolver method for finding cycles

On the occasion, the vertex is not in S_1 , we can test whether any neighbour of j (*i.e.*, root of $\phi_l(j, x)$) is in $\mathbb{F}_p$, without actually computing the roots of it (Line 15) by computing the gcd of two polynomials, as in the proof of Lemma 40. This will also require the same cost of $O(l^2 \log p)$. This incorporates the Supersolver strategy in this context.

After this, root computation is essential, and the best cost can be approximated by $O(l^2 \log p)$ (Line 24), with the roots being computed with probabilistic algorithms (Lemma 37). This cost is asymptotically equal to the cost in [EHL+20], with the gain being practical.

We are able to view a larger proportion of the graph at each step, which will provide better options to choose favourable routes to reach the frobenius conjugate earlier. The time estimates (in Section 5), we get after performing experiments, comparing our version with $i = 3$ with Algorithm 3.4 in [EHL+20] confirms to this gain.

5 Implementating conjugate path-finding algorithm with $i = 1$ and $i = 3$

We performed experiments with different primes p and small primes l , to find a paths to frobenius conjugate vertices by performing a pseudo-random walk in $\chi_l(\overline{\mathbb{F}_p})$. Our aim was to compare choice of $i = 3$ (Algorithm 1) with the adaptation of Supersolver method with $i = 1$ in Algorithm 3.4 of [EHL+20].

5.1 Set-up

We choose three primes 70001, 90001, 100003 (same as in [EHL+20] for good comparison), and for each we choose a certain number of supersingular j -invariants in $\mathbb{F}_{p^2}$. Our aim is to consider all primes l between 2 and 59 and for each prime l , we try to find a path to the conjugate of any given supersingular j invariant. We have recorded in Appendix C, the time for each p, l to find l^B isogenies for $B = 1, 3$ for all the chosen j -invariants. To show that the observations hold across all cases we have done 5 kinds of randomisations for each input j_0 . We have mentioned the total time in Table 1, for each case of randomisation⁹.

⁹For each of the cases, corresponding to various values of B , there are 5 entities of the form $(\dots)$.

Algorithm 1: Improved algorithm for finding conjugate paths.

Input: A prime p , a supersingular j -invariant $j_0 \in \mathbb{F}_{p^2}$ and a small prime $l(\neq p)$.

Output: A path P from j_0 to j_0^p in $\chi_l(\overline{\mathbb{F}_p})$.

```

1  $j \leftarrow j_0$ 
2  $P \leftarrow [j_0]$ .
3 while (Any of  $l^i, i = 0, 1, 2, 3$  isogeny is not found.) do
4   if  $j \in \mathbb{F}_p$  then
5     break //Note that  $l^0$ -isogeny is found between  $j$  and  $j^p$ .
6   else
7      $\bar{j} \leftarrow j^p$ .
8      $f \leftarrow \phi_l(j, x)$ .
9     if  $f(\bar{j}) = 0$  then
10      Append  $\bar{j}(= j^p)$  to  $P$ . //  $j^p$  is appended to the path.
11      break //Note that  $l^1$ -isogeny is found between  $j$  and  $j^p$ .
12   else
13      $g_1 = \frac{1}{2}[f + \pi(f)]$ .
14      $g_2 = \frac{-\beta}{2}[f - \pi(f)]$ .
15     if  $\text{InvFreeGCD}(g_1, g_2)$  then
16       Compute the intermediate root  $r_p \in \mathbb{F}_p$ .
17       Append  $r_p, \bar{j}(= j^p)$  to  $P$ .
18       break //Note that  $l^2$ -isogeny is found as  $j \rightarrow r_p \rightarrow j^p$ .
19   else
20      $j_1 \xleftarrow{R} \text{Roots}(f)$ . //Exclude the last root in P.
21     Append  $j_1$  to  $P$ .
22      $g_1 = \phi_l(j_1, x)$ .
23      $\bar{j}_1 = j_1^p$ .
24     if  $g_1(\bar{j}_1) = 0$  then
25       Append  $j_1, \bar{j}_1(= j_1^p), \bar{j}(= j^p)$  to  $P$ .
26       break // Note that  $l^3$ -isogeny is found as  $j \rightarrow j_1 \rightarrow j_1^p \rightarrow j^p$ .
27    $j \xleftarrow{R} \text{Roots}(g_1)$ .
28   Append  $j$  to  $P$ . //Exclude the last root in P.
29  $N := \#P - (i + 1)$  where  $i$  arises if  $l^i$  isogeny has occurred. //When we travel the path  $P$  from
    the last point, then  $P[N]$  is the first point whose frobenius conjugate is not in the path.
30  $P := P \cup \{P[i]^p : i \in [N \dots 1]\}$ . //Appending the frobenius conjugates in reverse way.
31 Return  $P$ .
```

Table 1: Table showing time required for each of the cases for different primes and values of $i = 1$ and $i = 3$.

p	Total supersingular j invariants	Time for $B = 1$	Time for $B = 3$	Ratio of speed-up
70001	3020	5221	2417	2.2
90001	10900	20700	10326	2
100003	22765	36778	20487	2

5.2 Observations

From the time comparison in Table 1, we see that path-finding to conjugate vertices speeds up when $i = 3$, compared to $i = 1$. The last column is the ratio of total time for $i = 1$ divided by the total time for $i = 3$, where the total takes into account all randomisations. This points out the factor by which it accelerates and we note that this happens uniformly across each of the primes, in each case of randomisation.

This means the experimental observations agree with the theoretical result that $i = 3$ is practically optimal when finding a path from any vertex to its Frobenius conjugate.

6 Choice of l

The number of supersingular elliptic curves in $S_{i(i \geq 1)}$ increases, in the graph $\chi_l(\overline{\mathbb{F}_p})$ on increasing l . By Lemma 6 of [CLG09] for mirror paths of length i , it is given by $l^{\frac{i}{2}} O(\sqrt{p})$. The number of isogeny cycles also increases with the increase in l [Theorem 7.1 of [ACL⁺24]]. This favour towards a higher value of l is also aggravated by the fact that using only $l = 2$ may result in some amount of non-randomness, short cycles, or walks.

However, the cost of root finding also increases with the increment of l . This means that to choose l to minimize time, balance is important.

We note from the experimental observations in Appendix C that the l for which the time is least to find a mirror path varies with p . We also performed experiments [Table 5] with other values of p and small values of l , where also the optimal l differed. These observations indicate that given p , we need to choose l after calculating the cost at each point.

6.1 Cost to optimize to pre-compute l in preprocessing phase

For a prime l , we can compute the cost per node, c_l by taking into account the exact costs of dominant multiplication operation in Algorithm 1. We can estimate the length L of a random walk from Proposition 22, and calculate the total cost $T_l = c_l \times L$. From estimations of cardinalities (or lower bounds) of the associated class group or using methods in literature [GJ16, MS24, G 18] to compute it, we can pre-compute the total cost T_{l_i} for $1 \leq i \leq t$, for some small primes $l_1 < l_2 < \dots < l_t$. We can

then minimize the function T_l with respect to l and choose $l = l_i$ when $T_{l_i} = \min(T_{l_1}, T_{l_2}, \dots, T_{l_i})$. We choose l , keeping in mind the splitting of p in an associated number field, so that short cycles [Section 5.3.4 [CLG09]] are not present.

7 Future Work

Several open directions arise from our results, namely: The need for tighter estimations of cardinalities of sets dealing with isogenies between conjugates, running-time trade-off analysis between curve exploration and isogeny computation, comparison with recent endomorphism ring algorithms; efficient membership testing without computing roots. Let us elaborate on these points:

- We have proved that to find path between conjugates in the graph $\chi_l(\overline{\mathbb{F}_p})$ by seeing l^i isogenies the optimal value of i is $i = 3$ and the optimal l depends on the cardinality of the target set $S_0 \cup S_1 \cup \overline{S_2} \cup \overline{S_3}$. There exists precise estimations of bounds on S_1 [Theorem 3.9 [EHL⁺20]] and S_0 [Equation 1, [DG16]]. This implies it remains a later work to derive estimations, especially the lower bounds on the cardinality of $\overline{S_i}$ up to $i \leq 3$.
- The exact value of the actual running time to find cycles depends on the trade-off between the number of curves checked and the cost of isogeny checks. This trade-off analysis is an open direction of future investigation.
- It would be also interesting to compare time estimates between the following to compute a endomorphism ring with cryptographically relevant parameters:
 1. Method of [EHL⁺20], with the least time required for various l when our method adapts Supersolver strategy [CRSCS22] to the sub-routine of conjugate path-finding.
 2. Strategy in the latest algorithms [FIK⁺23, ES24, XZQ24].

This will exemplify that any improvement of the isogeny computation problem is applicable to the equivalent problem of endomorphism ring computation and vice-versa.

- Another future scope of work is to test membership of elements in $\overline{S_3}$ without finding the roots exactly. If it leads to a speed-up, a similar approach for other $\overline{S_i}$ can also help in improving run-time by raising the value of i , which would aid in viewing a larger portion of the graph.

8 Conclusion

The problem of finding a path between supersingular elliptic curves which are Frobenius conjugates of each other, is a specialized case of the general path-finding problem and is closely connected to the problem of computing endomorphism rings. The hardness of the last two equivalent problems is a pillar of isogeny-based cryptography.

We prove that $i = 3$ is the optimal value to find mirror paths, by viewing i steps at a time, when we walk over the graph $\chi_l(\overline{\mathbb{F}_p})$, of all supersingular elliptic curves over $\mathbb{F}_{p^2}$ connected by l isogenies for given primes l, p . This balances the cost of computing isogenies with the advantage of larger overview,

a question raised in [EHL⁺20]¹⁰.

This implies that the relevant sets for mirror paths as required for Proposition 22, is $\bigcup_{i=1}^3 S_i$, where S_i is defined in equation 5. This observations opens the scopes of certain future research directions to estimate cardinalities and test membership in $S_i, i \leq 3$, and compute endomorphism rings with the improved versions.

We have adapted the Supersolver method [CRSCS22] in this problem of path-finding between conjugates. Our experiments confirms that $i = 3$ leads to time savings and reveals that l must be pre-computed to optimize time. These considerations on l and i will accelerate the process of pathfinding among Frobenius conjugate, which will result in a speed-up of cycle finding. Concrete application scenarios like general endomorphism ring computation [EHL⁺20], SQIsign [DFKL⁺20] or construction of collision resistant hash functions [CLG09] which depends on the hardness of cycle finding will benefit from this practical improvement.

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¹⁰Remark 3.5 speaks of the issue of bounding “by some fixed integer B ”

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A Experimental observations with small distance

Previous observations [ACNL⁺23] point out that paths between conjugate vertices are much more common than those between arbitrary vertices. In this section, we report the relative frequency of vertices in $\overline{S_2}, \overline{S_3}, \overline{S_4}$ from the experiments of performing pseudo-random walks.

We focussed on three primes p equal to 70001, 90001, 100003 and l ranging from primes 2 to 59. The difference from the earlier [ACNL⁺23] observations in literature is that, we consider a larger set for primes l , instead of only 2, 3 and deal with child nodes in $\mathbb{F}_p$ or S_1 . We note that precise bounds [Theorem 23, Theorem 24] on the estimate of cardinality of S_0, S_1 are known, whereas the same for S_i is absent in literature for $i > 1$. We considered the graph $\chi_l(\overline{\mathbb{F}_p})$, where we took random supersingular j -invariants, j_0 and continued pseudo-random walks by initially choosing $j = j_0$ and then j as a random root of $\phi_l(j, x)$.

The aim was to see when can we get a hint of a mirror path to j_0^p . This could happen when some intermediate j and j^p were connected by l^i isogeny. We varied $i = 2, 3, 4$ and stopped when a desired isogeny was found. We labeled each of the l roots (leaving out the root that has occurred previously) as $j_{i_1}, j_{i_2}, \dots, j_{i_l}$.

We halted when there was some $i_1, i_2 \in \{1, 2, \dots, l\}$ such that either $j_{i_1} = j_{i_2}^p$, or $\phi_l(j_{i_1}, j_{i_2}^p) = 0$ or $\deg(\gcd(\phi_l(j_{i_1}, x), \phi_l(x, j_{i_2}^p))) \geq 1$. This corresponds to the cases where $j \in S_2$ or S_3 or S_4 respectively, j being the parent of j_{i_1} and j_{i_2} . When $i_1 = i_2$, the child $j_{i_1} \in S_0$ (for $i = 2, 4$) or S_1 (for $i = 3$). These lead to mirror paths that can be utilised to construct cycles. Contrarily when $i_1 \neq i_2$, the paths lead to short cycles which are discarded from the viewpoint of cryptographic security.

In Tables 2, 3, 4, we have noted l, p along with the cases for which we get mirror paths when we perform pseudo-random walks initiating from an arbitrary supersingular j invariant in $\mathbb{F}_{p^2}$. We have mentioned the total number of such j -invariants in the caption of each table. The percentage of points in $\overline{S_i}$ is in bold digits, which we have calculated using the total j -invariants from which we initiate. For each prime and each isogeny l , we have noted the number of elements in S_i and $\overline{S_i}$ for $i = 2, 3, 4$. Clearly, we can easily infer that the number of points in $S_i \setminus \overline{S_i}$ is quite small. This means that the number of supersingular j invariants such that j_{i_1} and $j_{i_2}^p$ are isogeneous for $i_1 \neq i_2$ is quite less in general. This agrees with the previous theoretical analysis in [ACNL⁺23]. This practical idea of the comparison of the proportional cardinalities is new and is a step further towards studying the distances between conjugates and arbitrary vertices as was suggested in Section 4.1 of [ACNL⁺23].

This once again reassures that paths between conjugate j -invariants are much more common than any other paths. Along with that, it points to the new direction that paths arising due to vertices in $\overline{S_2}, \overline{S_3}, \overline{S_4}$ are quite common and so efficient detection strategies for $i \geq 2$ would be helpful for obtaining mirror paths.

We have experimentally studied the issue of mirror paths arising from vertices in S_2, S_3, S_4 for various values of l . We see that the derivation of better theoretical estimates of the cardinality of sets $\overline{S_i}$ is important from the viewpoint of obtaining paths between conjugate vertices in the supersingular isogeny graph.

Also, this suggests searching for optimal distance and better l to find mirror paths. We have done this in Sections 4.1 and 6 respectively.

Table 2: Analysis of conjugate paths and detection of l^i isogenies for small i for $p = 70001$ and total of 604 supersingular j invariants. Percentage of the cardinality of $\overline{S}_i$ for $i = 2, 3, 4$ is printed in bold.

l	$\#S_2$	$\#\overline{S}_2$	$\frac{\#S_2 \times 100}{Total}$	$\#S_3$	$\#\overline{S}_3$	$\frac{\#S_3 \times 100}{Total}$	$\#S_4$	$\#\overline{S}_4$	$\frac{\#S_4 \times 100}{Total}$
2	32	32	6	89	80	14	483	483	80
3	30	30	5	266	253	42	308	308	51
5	66	66	11	171	142	24	367	367	61
7	110	110	19	274	246	41	220	220	37
11	146	146	25	199	179	30	259	259	43
13	170	170	29	276	241	40	158	126	21
17	237	237	40	154	147	25	213	119	20
19	247	247	41	349	300	50	8	3	1
23	287	287	48	285	262	44	32	17	3
29	291	291	49	291	181	30	22	13	3
31	316	316	53	288	205	34	0	0	0
37	370	370	62	234	112	19	0	0	0
41	410	410	68	194	96	16	0	0	0
43	417	417	70	187	92	16	0	0	0
47	452	452	75	152	61	11	0	0	0
53	470	470	78	134	20	4	0	0	0
59	484	484	81	120	29	5	0	0	0

B Algorithm to test for vertices in S_1 without computing roots

C Time to find conjugate path for various values of p and l

In this section we record the time to find path to conjugate vertex for each prime p, l . We represent each as a size-18 tuple of the form $(t_1, t_2, \dots, t_{17}; T)$, where the first 17 entries correspond to the time for each $l \in [2, 59]$ consecutively, and the right-most entry T denotes the total time (which includes a small overhead cost). The times listed in Table 1 represent the sum of the total timings T for each value of B .

C.1 Time when $p = 70001$

C.1.1 $B = 1$

(10, 8, 16, 8, 29, 23, 51, 18, 50, 108, 23, 87, 121, 109, 117, 217, 135; 1138), (11, 7, 14, 8, 25, 23, 49, 17, 34, 65, 14, 61, 81, 77, 86, 202, 166; 950), (10, 7, 13, 11, 29, 27, 54, 18, 36, 67, 13, 63, 85, 99, 111, 233, 149; 1033), (10, 9, 18, 10, 32, 36, 72, 26, 60, 100, 23, 100, 136, 94, 73, 156, 109; 1072), (10, 8, 15, 11, 26, 21, 47, 17, 35, 77, 15, 68, 84, 75, 79, 232, 199; 1028).

Table 3: Analysis of conjugate paths and detection of l^i isogenies for small i for $p = 90001$ and total of 2180 supersingular j invariants. Percentage of the cardinality of $\bar{S}_i$ for $i = 2, 3, 4$ is printed in bold.

l	$\#S_2$	$\#\bar{S}_2$	$\frac{\#S_2 \times 100}{Total}$	$\#S_3$	$\#\bar{S}_3$	$\frac{\#S_3 \times 100}{Total}$	$\#S_4$	$\#\bar{S}_4$	$\frac{\#S_4 \times 100}{Total}$
2	41	41	2	1176	1143	53	963	963	45
3	67	67	4	1118	1075	50	995	995	46
5	65	65	3	1384	1311	61	731	731	34
7	152	152	7	953	883	41	1075	1075	50
11	203	203	10	1643	1454	67	334	334	16
13	246	246	12	746	688	32	1188	924	43
17	342	342	16	1416	1274	59	422	216	10
19	400	400	19	1414	1276	59	366	85	4
23	436	436	20	1673	1517	70	71	17	1
29	513	513	24	1625	1286	59	42	6	1
31	561	561	26	1617	1216	56	2	1	1
37	661	661	31	1503	751	35	16	5	1
41	705	705	33	1475	1030	48	0	0	0
43	662	662	31	1500	588	27	18	4	1
47	705	705	33	1475	958	44	0	0	0
53	810	810	38	1370	352	17	0	0	0
59	889	889	41	1291	714	33	0	0	0

C.1.2 $B = 3$

(9, 7, 9, 8, 12, 11, 16, 12, 16, 27, 14, 27, 45, 43, 49, 77, 76; 466), (8, 6, 8, 6, 11, 10, 14, 9, 16, 25, 11, 23, 37, 39, 47, 70, 71; 420), (14, 8, 14, 9, 14, 15, 20, 13, 23, 38, 18, 39, 54, 50, 57, 91, 77; 564), (11, 10, 9, 7, 13, 10, 22, 12, 18, 23, 12, 32, 53, 53, 50, 87, 85; 517), (10, 8, 10, 6, 10, 13, 17, 10, 17, 27, 12, 27, 42, 40, 49, 74, 71; 450).

C.2 Time for $p = 90001$

C.2.1 $B = 1$

(59, 59, 65, 104, 69, 278, 166, 184, 141, 175, 103, 489, 266, 671, 137, 529, 220; 3724), (54, 48, 60, 91, 64, 289, 182, 186, 169, 238, 147, 502, 333, 1114, 213, 548, 184; 4431), (44, 42, 51, 81, 51, 203, 170, 226, 192, 226, 141, 505, 273, 844, 179, 489, 175; 3900), (71, 72, 76, 120, 76, 238, 121, 147, 116, 272, 167, 426, 221, 1016, 199, 682, 245; 4277), (55, 52, 56, 99, 67, 274, 167, 201, 168, 238, 145, 518, 269, 879, 202, 696, 270; 4368).

C.2.2 $B = 3$

(56, 51, 59, 72, 61, 135, 96, 118, 103, 138, 98, 183, 110, 206, 99, 322, 183; 2100), (70, 52, 70, 88, 62, 151, 75, 81, 75, 98, 71, 148, 160, 325, 128, 193, 122; 1979), (68, 60, 65, 90, 67, 155, 74, 78, 73, 97, 75, 150, 161, 350, 116, 195, 120; 2004), (62, 50, 64, 73, 55, 133, 104, 113, 101, 136, 100, 178, 111, 202, 101, 320, 174; 2084), (43, 37, 44, 52, 40, 96, 73, 78, 98, 159, 123, 246, 143, 348, 137, 248, 184; 2159).

Table 4: Analysis of conjugate paths and detection of l^i isogenies for small i for $p = 100003$ and total of 4553 supersingular j invariants. Percentage of the cardinality of $\bar{S}_i$ for $i = 2, 3, 4$ is printed in bold.

l	$\#S_2$	$\#\bar{S}_2$	$\frac{\#S_2 \times 100}{Total}$	$\#S_3$	$\#\bar{S}_3$	$\frac{\#S_3 \times 100}{Total}$	$\#S_4$	$\#\bar{S}_4$	$\frac{\#S_4 \times 100}{Total}$
2	78	78	2	3286	3169	70	1189	1189	27
3	160	160	4	2123	2026	45	2270	2270	50
5	257	257	6	3223	2899	64	1073	1073	24
7	350	350	8	1201	1131	25	3002	3002	66
11	421	421	10	2343	2186	49	1789	1789	40
13	542	542	12	3285	2898	64	726	543	12
17	601	601	14	3462	3085	68	490	225	5
19	775	775	18	1998	1847	41	1780	603	14
23	862	862	19	3292	3007	67	399	98	3
29	1112	1112	25	3327	2986	66	114	40	1
31	1198	1198	27	3189	2198	49	166	47	2
37	1383	1383	31	3166	1906	42	4	0	0
41	1435	1435	32	3116	2385	53	2	0	0
43	1508	1508	34	3039	894	20	6	0	0
47	1694	1694	38	2859	1784	40	0	0	0
53	1779	1779	40	2774	1726	38	0	0	0
59	1969	1969	44	2584	1010	23	0	0	0

C.3 Time for $p = 100003$

C.3.1 $B = 1$

(96, 140, 76, 341, 247, 131, 267, 680, 454, 506, 815, 527, 449, 1096, 373, 346, 863; 7420), (90, 143, 83, 343, 249, 132, 274, 670, 462, 514, 827, 533, 459, 1098, 374, 425, 808; 7493), (63, 89, 53, 271, 393, 199, 235, 729, 397, 446, 654, 367, 463, 1733, 480, 314, 662; 7557), (61, 88, 53, 289, 370, 208, 226, 712, 437, 466, 647, 370, 464, 1548, 516, 319, 649; 7434), (93, 134, 74, 379, 324, 156, 179, 727, 322, 333, 595, 432, 486, 1166, 404, 352, 709; 6874).

C.3.2 $B = 3$

(92, 106, 60, 200, 202, 133, 176, 251, 183, 208, 410, 231, 272, 389, 236, 269, 427; 3854), (91, 104, 65, 208, 210, 128, 163, 247, 186, 204, 411, 240, 261, 382, 228, 274, 418; 3828), (79, 93, 55, 186, 195, 134, 184, 303, 277, 263, 360, 298, 371, 528, 315, 292, 453; 4397), (64, 76, 48, 146, 156, 156, 202, 320, 217, 240, 368, 298, 316, 538, 238, 231, 371; 3995), (101, 115, 71, 227, 148, 97, 125, 229, 302, 259, 255, 262, 372, 597, 372, 356, 514; 4413).

D Value of l to optimize time

We choose primes of size 35 to 45 bits and for each prime, we generated a list of supersingular j -invariants randomly. We performed a pseudo-random walk for each j -invariant until we found a mirror

Algorithm 2: Testing for vertices in S_1 .

Input: A prime l , a positive integer d and a supersingular j -invariant $j_{00} \in \mathbb{F}_{p^2}$.

Output: Whether there exists a vertex $j_d \in S_1$ such that j_{00} and j_d are l^d -isogeneous.

```
1  $M \leftarrow m_d(j_{00}, a + \beta b)$ . // Bivariate polynomial in  $\mathbb{F}_{p^2}[a, b]$ .
2  $\Phi \leftarrow \phi_l(a + \beta b, a - \beta b)$ . // Bivariate polynomial in  $\mathbb{F}_p[a, b]$ .
3  $R = \text{Resultant}(M, \Phi, a)$ . // Resultant of above polynomials with respect to  $a$ .
4 if  $R$  has a root  $b_0$  in  $\mathbb{F}_p$  then
5    $g = \text{GCD}(M(a, b_0), \Phi(a, b_0))$ .
6   if  $g$  has a root  $b_0$  in  $\mathbb{F}_p$  then
7     Return Yes.
8 Return False.
```

path, in $\chi_l(\overline{\mathbb{F}_p})$ for primes $l \in [2, 59]$ ¹¹.

We noted the optimal l in Table 5 which balances the costs with number of roots. The difference in time required by various values of l should increase when larger number of cycles have to be constructed in practical situations.

Table 5: Table showing that for a given prime, there is a prime l , which finds a requisite vertex for mirror path in shorter time.

p	Best l
34359738337	7 and 3
274877906857	7
1099511627689	3
2199023255521	13
8796093022141	17
35184372088777	23

E Codes for each part

Our github link mentioned in Section 1.1 corresponds to the codes and corresponding output files for the following:

- The folder “folder_experiments_small_distance” corresponds to results in Section 5 and Section C.
- The folder “folder_experiments_favourable_l” corresponds to results in Section D.
- The folder “sets_Si” corresponds to results in Section A.

¹¹The total number of vertices through which we walk are substantial enough for inference and can be checked from our code in GitHub.